

Debugging

Overview

- What is a bug?
- What is debugging?
- Debugging with VS Code

Learning Objectives

- Identify what debugging is
- Explain how to debug in VS Code

What is a bug?

Generation

- Humans make mistakes, when we make mistakes in code they're called bugs 🐛
- There are many types of bugs. Common causes could be an accidental mistake, a misunderstanding of how something works or a flaw in the logic of the code.
- Bugs can be fairly benign or they can cause planes to drop out the sky and kill people

```
# Accidental mistake  
prnt("Hello buggy software!")
```

```
# Misunderstanding of how something works  
drinks_list = []  
drinks_list = drinks_list.append("Coke")
```

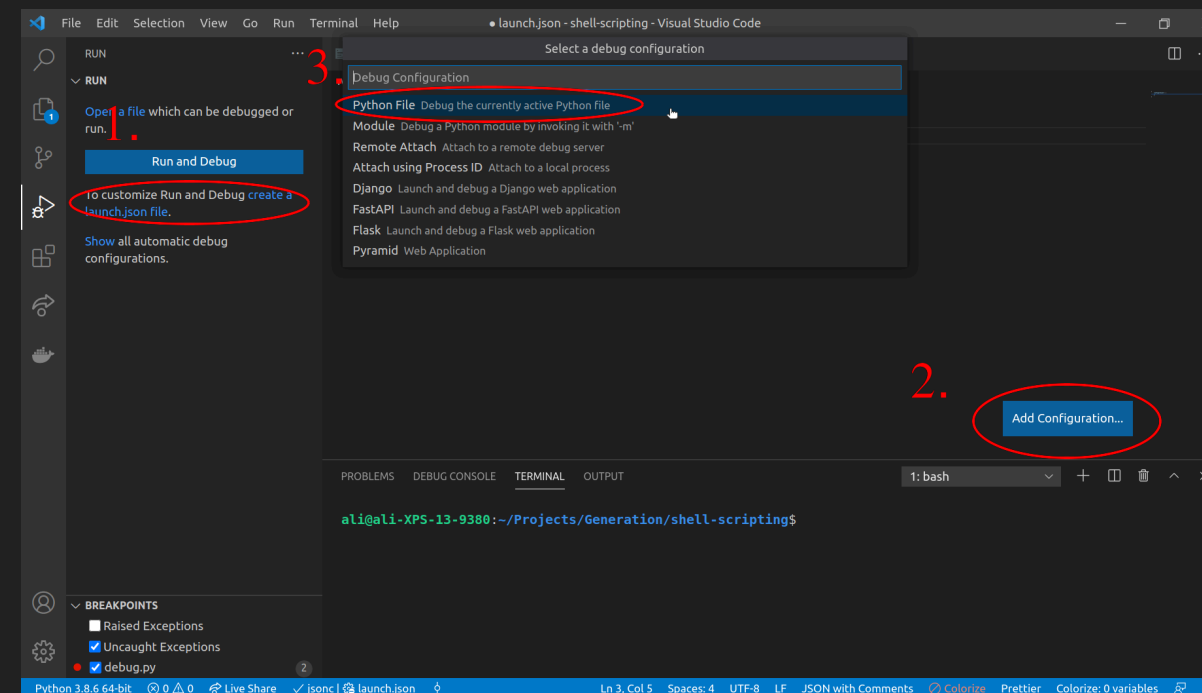
```
# Flawed logic  
age = int(input("What is your age?"))  
minimum_drinking_age = 18  
challenge 25 age = 25
```

What is Debugging?

- Debugging is the process of removing bugs from your code
- You can try and find / isolate bugs in your code using debugging tools
- VS Code is capable of running your code line by line and displaying values of variables to you
- This can give you context for a bug and helps isolate the problem

Before you can debug you need to configure your debugger. Luckily VS Code makes this super easy (as usual!)

1. Open the debugging Window **Ctrl + Shift + D**
2. Create a launch.json file
3. Add a configuration
4. Select Python from the dropdown



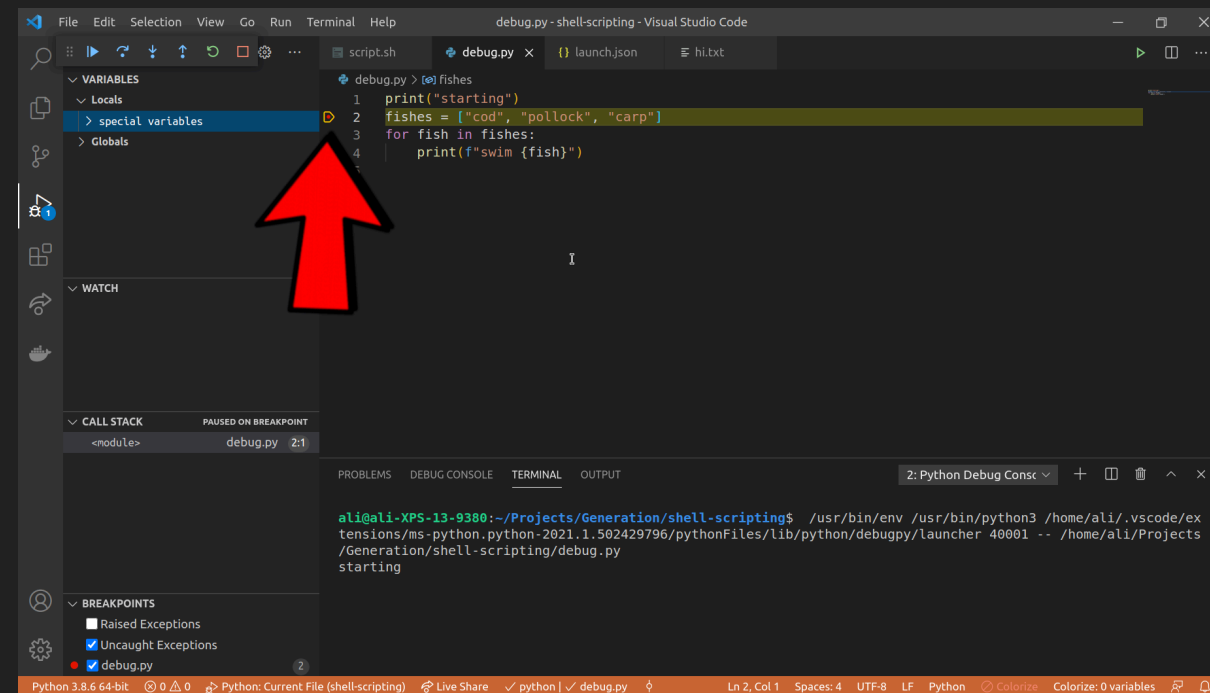
Launch Config File

This will create a `launch.json` file which contains the configuration for debugging.

```
{
  "configurations": [
    {
      "name": "Python: Current File",
      "type": "python",
      "request": "launch",
      "program": "${file}",
      "console": "integratedTerminal"
    }
  ]
}
```

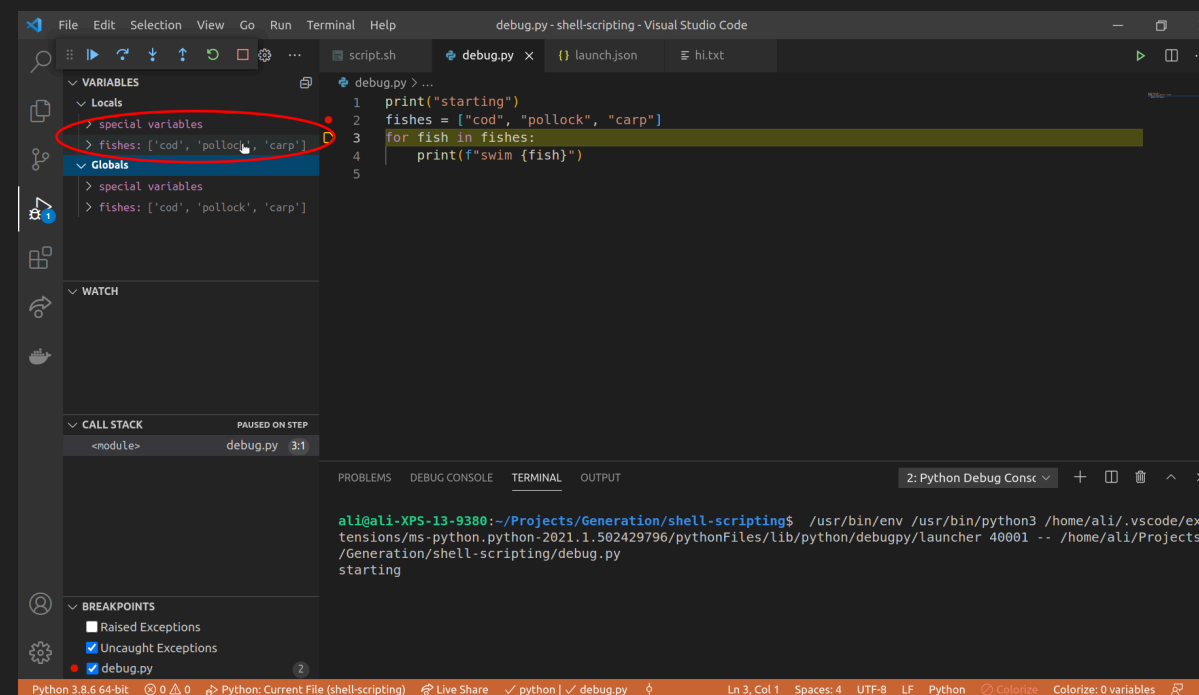

Let's Squash Some Bugs

1. Add a break point by clicking on the line
2. Run the debug configuration by click the green arrow in the debug VS Code tab or on Windows and Linux by pressing **F5**
3. Use the floating debug panel to control the debugger



Variables

- You can view the values of your local and global variables in the left hand panel
- You can update the values of these variables as you debug



Exercise

Setup debugging and debug `buggy_restaurant.py`

Overview - recap

- What is a bug?
- What is debugging?
- Debugging with VS Code

Learning Objectives - recap

- Identify what debugging is
- Explain how to debug in VS Code

Further Reading

- [VS Code Debugging Python](#)

Emoji Check:

On a high level, do you think you understand the main concepts of this session? Say so if not!

1. 🥲 Haven't a clue, please help!
2. 😞 I'm starting to get it but need to go over some of it please
3. 😐 Ok. With a bit of help and practice, yes
4. 😊 Yes, with team collaboration could try it
5. 😄 Yes, enough to start working on it collaboratively